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Hand-eye calibration method for a line structured light robot vision system based on a single planar constraint

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ABSTRACT

Hand-eye calibration is a prerequisite for robot vision system applications. However, due to the lack of image features, implementing hand-eye calibration with a line-structured light sensor is limited by complex procedures and special reference objects. The aim of this paper is to develop a stable and undemanding eye-in-hand calibration method with a 2D laser sensor that can be adapted to random measurement strategies in most common scenes. The proposed method can use an easily accessible plane from machined workpieces, except for specialized calibration objects, which facilitates the automation of industrial robots. Moreover, in this method, a two-step iterative method is combined with fast simulated annealing based on a single planar constraint to overcome large initial deviations and sensor errors. Simulations and calibration experiments are conducted to assess the method performance and demonstrate the feasibility and accuracy of the proposed eye-in-hand calibration method.

1. Introduction

Line-structured light robot vision systems are widely used in industrial scenarios. The participation of line structured light, namely, a 2D laser sensor, complements the perception of depth dimensions in monocular vision systems and increases the precision of measurements. Most of robot vision system applications rely on high-precision calibration of the visual perception chain, especially hand-eye calibration. Hand-eye calibration provides the relationship between the robot end effector and vision sensor [1], which is an active area of research for last three decades. The applications of robot vision can be divided into eye-to-hand and eye-in-hand configurations according to the installation position of the sensors [1]. In addition, in terms of application, an eye-to-hand configuration decreases the load on the robot arm to obtain a more stable perceptual scene and adapt to a more complex vision system. In contrast, the eye-in-hand configuration can provide a larger working space and more flexible operation modes due to its compact mechanical structure, but it requires a smaller and lighter sensor. The focus of this paper is on the development of an eye-in-hand line structured light robot vision system (Fig. 1c). This system can be used to improve the adaptability and ease of use of the calibration method.

Most of the hand-eye calibration methods for line structured light are similar to traditional hand-eye calibration methods. These methods

can be implemented to operate the robot in different poses to measure a fixed object with high manufacturing accuracy. The obtained image feature and robot pose data are used to solve the hand-eye calibration equation in the form of " $AX = XB$ " [1]. This means that when the robot end frames perform rigid motion A , the vision sensor frames experience motion B . X represents the unknown transformation from the vision sensor frame to the robot end frame. Separate methods [2–4] and simultaneous methods [5] are normally used to solved the rotation part and translation part of the unknown relation X , and there are also other researchers work for " $AX = YB$ " solution [6,7]. The line-structured light can be used to improve the measurement quality. However, it also introduces other problems into hand-eye calibration with a 2D laser profile. Compared with monocular vision, line structured light obviously reduces the amount of information obtained from images. Although depth information can be obtained, all the known points are concentrated on the same line in each image, which limits the use of the traditional hand-eye calibration solution.

Robot pose constraints and image feature constraints are two typical ideas that aim to provide additional conditions for solving the hand-eye calibration problem. Several researchers [8,9] have used robot pose constraints to assist in traditional hand-eye calibration. The pose constraint is used to simplify the calibration calculation to avoid complex optimizations. However, the robot pose or motion strategy for

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object measurement needs to be designed in advance, which adds an additional step involving error. With inappropriate poses or motion, operations with insufficient precision always lead to worse results. The image feature constraint was originally used to determine the relative position of a camera to an object. Several researchers [10–15] have designed special objects with 3D structural features to obtain image feature constraints. When the laser plane intersects these objects, more information is reflected in the images, which can further constrain the relationship between the camera and objects. The difficulty of high-precision machining of special calibration objects is the main obstacle to its widespread use. Two-dimensional patterns [16] are also sufficient for traditional hand-eye calibration to uniquely determine the relative position, but it becomes infeasible to implement line-structured light hand-eye calibration because the patterns are invisible in stripe images. And there are other works achieve hand-eye calibration based on 3D feature constraints using point clouds reconstruction. Paper [17] adopts 3D camera measuring 3D point clouds of daily objects to perform calibration, and paper [18] uses monocular camera for calibration with natural scene.

In recent years, several researchers have attempted to enhance the applicability of line-structured light hand-eye calibration by eliminating dedicated calibration objects. The idea is to abstract effective geometric features from traditional calibration objects and research the interaction of the laser plane and the geometric features that are visible in the image. The combination of robot pose constraints and image feature constraints is a common way to achieve this goal and has been applied in eye-to-hand configurations. In [19,20], the robot ends perform multiple sets of translational movements, and multiple sets of parallel line stripes on the objective plane are detected by a stationary camera to form a closed-form solution for the calibration problem. Ref. [21] focused on planar constraints and presented an iterative method for solving the eye-to-hand calibration problem. In this method, the points that lie on the plane $z = 0$ in the laser scanner coordinate system are considered, and equations are derived according to the planar constraints. More than three nonparallel planes are used for data collection, which is inconvenient in many scenarios. Additionally, as shown in [22], this method is sensitive to the selection of plane orientation angles, which increases the restrictions on the implementation of the method. [22] used a similar method but focused on a calibration strategy to simplify the data collection process. The robot poses are designed to detect special distributions of laser stripes, and semi-automatic eye-to-hand calibration is achieved on a single plane. Compared with the eye-to-hand calibration problem, little work has been devoted to solving the eye-in-hand problem with simple geometric features. [8] proposed a method based on detecting straight edges and multiple groups of translational movements, and the geometric constraint of straight lines was used to construct a two-step least-squares problem to solve the eye-to-hand calibration problem.

In the aforementioned calibration methods, complex measurement strategies and special calibration objects generally make the calibration inefficient and difficult to apply. In this paper, a line-structured light eye-in-hand calibration method is implemented with strong applicability using only a single planar reference. There are two main features in the proposed method:

- Only a single planar reference is needed: A two-step iterative method combined with simulated annealing (SA) is proposed to achieve a stable line-structured light eye-in-hand calibration solution. The preconditions of the method are just an initial estimate of the unknown result with large errors and a single planar reference from common machining workpieces placed arbitrarily.
- No measurement strategy limitations exist: The calibration can be implemented without any special measurement strategy because the established plane constraint equations apply to arbitrarily acquired points on the stripes. In other words, random measurements can lead to successful calibration, which is conducive to achieving an automatic calibration process.

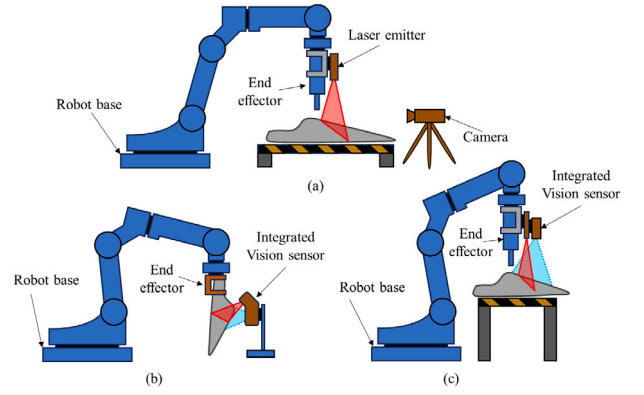


Fig. 1. Different configurations of the line-structured light robot vision system. (a) and (b) are two classical eye-to-hand configurations. (c) is a common eye-in-hand configuration that defines the calibration problem in this paper.

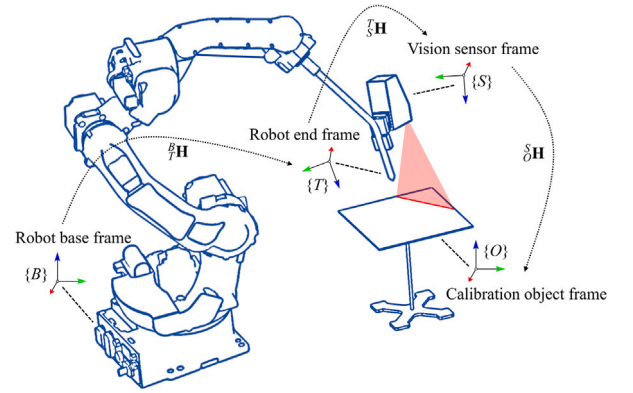


Fig. 2. The line-structured light robot vision system setup.

Based on these features, the proposed hand-eye calibration method can be implemented directly on workpieces with machined flat areas, which contributes to process simplification and precision maintenance for robotic manufacturing.

2. Methodology

2.1. Problem description

The eye-in-hand structured light hand-eye calibration scenario addressed in this paper is shown in Fig. 2, in which a vision sensor integrated with a monocular camera and a laser emitter is firmly attached to the end of the robot. The pattern projected on a plane is a straight line, and it is captured by a camera for measurement.

There are four coordinated systems defined in this scenario, as shown in Fig. 2: the robot base coordinated system $\{B\}$, the robot end coordinated system $\{T\}$, the vision sensor coordinated system $\{S\}$ and the calibration object coordinated system $\{O\}$. During the calibration, the robot base and the object plane are fixed to the ground. Thus, $\{B\}$ and $\{O\}$ can be regarded as coincident. Here, and hereinafter, the homogeneous transformation matrix $\mathbf{H}$ is used to define the transformation of the coordinate systems. For a brief description, a special nomenclature is used, as shown in Fig. 2. The form ${}^M_N\mathbf{H}$ describes the homogeneous transformation matrix of the coordinate system $\{N\}$ with respect to the coordinate system $\{M\}$. The homogeneous transformation matrix $\mathbf{H}$ can be divided into two parts: the rotation part $\mathbf{R}_H$ and the translation part $\mathbf{t}_H$. For example, $\mathbf{R}_{T^B}^S$ represents the rotation part

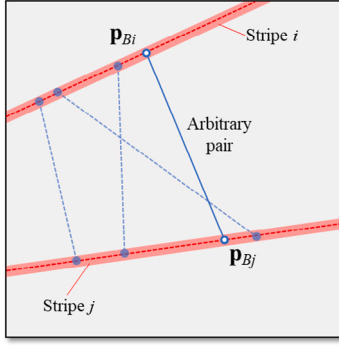


Fig. 3. Projected pattern from an arbitrary pair of measuring poses marked as Poses i and j .

of the transformation of the robot base coordinated system w.r.t the robot end coordinated system in the i th pose situation:

$${}^B_T H_i = \begin{bmatrix} \mathbf{R}_{T H_i}^B & \mathbf{t}_{T H_i}^B \\ 0 & 1 \end{bmatrix} \quad (1)$$

Considering the arbitrary point $\mathbf{p}_B$ that lies on the object plane in the robot base coordinate system, it has a relationship with its relative point $\mathbf{p}_S$ in the sensor coordinate system:

$$\begin{bmatrix} \mathbf{p}_B \\ 1 \end{bmatrix} = {}^B_T H_S^T \begin{bmatrix} \mathbf{p}_S \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{T H}^B & \mathbf{t}_{T H}^B \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{T H}^S & \mathbf{t}_{T H}^S \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{p}_S \\ 1 \end{bmatrix} \quad (2)$$

2.2. Two-step iterative method with planar references

Two-step methods are typical ways to solve the hand-eye calibration problem in a simple and efficient manner [1]. In this study, the calibration problem with planar constraints is analysed, and a separable solution for the rotation and translation parts, which is capable of handling poor features from simple references, is obtained. Furthermore, an iterative process, in which the rotation and translation parts are sequentially updated, is employed to mitigate the accumulation of errors and enhance the solution stability.

First, planar constraints assist in rebuilding hand-eye problems with poor features in stripe images. Studying the relationship between measuring pose pairs is a common way to solve traditional hand-eye calibration problems. Fig. 3 shows the projected stripe pattern from an arbitrary pair of measuring poses marked as Poses i and j . From Eq. (2), any point on the laser stripe satisfies the following:

$$\mathbf{p}_{Bi} = \mathbf{R}_{T H_i}^B \mathbf{R}_{S H}^T \mathbf{p}_{Si} + \mathbf{R}_{T H_i}^B \mathbf{t}_{S H}^T + \mathbf{t}_{T H_i}^B \quad (3)$$

$$\mathbf{p}_{Bj} = \mathbf{R}_{T H_j}^B \mathbf{R}_{S H}^T \mathbf{p}_{Sj} + \mathbf{R}_{T H_j}^B \mathbf{t}_{S H}^T + \mathbf{t}_{T H_j}^B \quad (4)$$

The vector $\mathbf{u}_{Bij}$ that connects two laser stripes in the robot base coordinate system can be represented as follows:

$$\mathbf{u}_{Bij} = \mathbf{p}_{Bi} - \mathbf{p}_{Bj} \quad (5)$$

Then, because of the planar constraint, it is obvious that:

$$\mathbf{n}_B^T \mathbf{u}_{Bij} = 0 \quad (6)$$

where $\mathbf{n}_B^T$ represents the normal vector to the calibration plane. Thus, we can rebuild the hand-eye calibration problem as follows:

$$\arg \min_{{}^T_S H} \sum_{i,j} \|\mathbf{n}_B^T \mathbf{u}_{Bij}\| \quad (7)$$

It is obvious that $\mathbf{n}_B^T$ cannot be obtained without additional measurements. However, utilizing an iterative method is helpful for gradually approaching this value. Before the calibration process, an initial estimate of ${}^T_S H$ should be obtained, which allows a large margin of

error. The initial estimate $\widehat{{}^T_S H}$ can be the result of the last rough calibration, the nominal dimensions on design drawings or even the estimation by manual measurement. Then, the estimated point cloud $\{\widehat{\mathbf{p}}_B\}$ can be generated by Eq. (2). According to the coplanar constraint, the normal vector can be estimated by solving the following equation:

$$\begin{bmatrix} \widehat{\mathbf{p}}_{B1}^T \\ \widehat{\mathbf{p}}_{B2}^T \\ \widehat{\mathbf{p}}_{B3}^T \\ \vdots \end{bmatrix} \widehat{\mathbf{n}}_B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \end{bmatrix} \quad (8)$$

Additionally, $\widehat{\mathbf{n}}_B$ can be solved by principal component analysis (PCA).

Then, we discuss the solution of Eq. (7), which can be expanded as follows:

$$\arg \min_{{}^T_S H} \sum_{i,j} \left\| \widehat{\mathbf{n}}_B^T (\mathbf{R}_{T H_i}^B \mathbf{R}_{S H}^T \mathbf{p}_{Si} - \mathbf{R}_{T H_j}^B \mathbf{R}_{S H}^T \mathbf{p}_{Sj}) + \widehat{\mathbf{n}}_B^T (\mathbf{R}_{T H_i}^B - \mathbf{R}_{T H_j}^B) \mathbf{t}_{S H}^T + \widehat{\mathbf{n}}_B^T (\mathbf{t}_{T H_i}^B - \mathbf{t}_{T H_j}^B) \right\| \quad (9)$$

The Kronecker product and matrix vectorization are utilized for variable separation and can be defined as follows:

$$\mathbf{A}_{m \times n} \otimes \mathbf{B} = \begin{bmatrix} a_{11} \mathbf{B} & \dots & a_{1n} \mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1} \mathbf{B} & \dots & a_{mn} \mathbf{B} \end{bmatrix} \quad (10)$$

$$\text{Vec}(\mathbf{A}_{m \times n}) = [a_{11} a_{21} \dots a_{m1}; \dots; a_{1n} a_{2n} \dots a_{mn}]^T \quad (11)$$

Thus, the first part of Eq. (9) can be rewritten as follows:

$$\begin{aligned} & \widehat{\mathbf{n}}_B^T (\mathbf{R}_{T H_i}^B \mathbf{R}_{S H}^T \mathbf{p}_{Si} - \mathbf{R}_{T H_j}^B \mathbf{R}_{S H}^T \mathbf{p}_{Sj}) \\ &= \widehat{\mathbf{n}}_B^T (\mathbf{p}_{Si}^T \otimes \mathbf{R}_{T H_i}^B - \mathbf{p}_{Sj}^T \otimes \mathbf{R}_{T H_j}^B) \mathbf{r}_{S H}^T \end{aligned} \quad (12)$$

where

$$\mathbf{r}_{S H}^T = \text{Vec}(\mathbf{R}_{T H}^S)$$

On this basis, a brief form of the hand-eye calibration problem can be obtained as follows:

$$\arg \min_{\mathbf{r}_{S H}^T, \mathbf{t}_{S H}^T} \sum_{i,j} \| \mathbf{C}_1 \mathbf{r}_{S H}^T + \mathbf{C}_2 \mathbf{t}_{S H}^T + d_1 \| \quad (13)$$

where

$$\mathbf{C}_1 = \widehat{\mathbf{n}}_B^T (\mathbf{p}_{Si}^T \otimes \mathbf{R}_{T H_i}^B - \mathbf{p}_{Sj}^T \otimes \mathbf{R}_{T H_j}^B)$$

$$\mathbf{C}_2 = \widehat{\mathbf{n}}_B^T (\mathbf{R}_{T H_i}^B - \mathbf{R}_{T H_j}^B)$$

$$d_1 = \widehat{\mathbf{n}}_B^T (\mathbf{t}_{T H_i}^B - \mathbf{t}_{T H_j}^B)$$

It is difficult to solve for the rotation and translation parts simultaneously, and separate methods [8,16] can reduce the complexity of the solution and overcome the lack of image features. Thus, a two-step method is utilized to solve for the rotation part first.

In other words, the unknown transform part in Eq. (13) is replaced by an estimate value. Then, this equation becomes a representation of the least squares problem.

$$\mathbf{r}_{S H}^{*T} = \arg \min_{\mathbf{r}_{S H}^T} \sum_{i,j} \| \mathbf{C}_1 \mathbf{r}_{S H}^T + \mathbf{C}_2 \widehat{\mathbf{t}}_{S H}^T + d_1 \| \quad (14)$$

Obviously, the operation $\mathbf{R}_{S H}^{*T} = \text{Vec}^{-1}(\mathbf{r}_{S H}^{*T})$ cannot guarantee that $\mathbf{R}_{S H}^{*T}$ is an orthogonal matrix because of the estimate error. Thus, the nearest proper orthogonal matrix to $\mathbf{R}_{S H}^{*T}$ should be found to update the estimate of the rotation matrix. Singular value decomposition (SVD) is an efficient method for handling problems with good approximation ability [23]. If $\mathbf{R}_{S H}^{*T} = \mathbf{U} \Sigma \mathbf{V}^T$ is obtained via SVD, the nearest rotation matrix can be obtained as follows:

$$\mathbf{R}_{S H}^{*T} = \begin{cases} \mathbf{U} \mathbf{V}^T & \det(\mathbf{R}_{S H}^{*T}) \geq 0 \\ \mathbf{U} \text{diag}(1, 1, -1) \mathbf{V}^T & \det(\mathbf{R}_{S H}^{*T}) < 0 \end{cases} \quad (15)$$

Afterward, the translation part can be obtained similarly by the following equation:

$$\mathbf{t}_{S\mathbf{H}}^* = \arg \min_{\mathbf{t}_{S\mathbf{H}}} \sum_{i,j} \|\mathbf{C}_2 \mathbf{t}_{S\mathbf{H}} + d_2\| \quad (16)$$

where

$$d_2 = \widehat{\mathbf{n}}_B^T (\mathbf{R}_{T\mathbf{H}_i}^B \mathbf{R}_{S\mathbf{H}}^\dagger \mathbf{p}_{Si} - \mathbf{R}_{T\mathbf{H}_j}^B \mathbf{R}_{S\mathbf{H}}^\dagger \mathbf{p}_{Sj} + \mathbf{t}_{T\mathbf{H}_i} - \mathbf{t}_{T\mathbf{H}_j})$$

2.3. Solution analysis

To ensure the stability of the solution, it is necessary to consider the physical significance of Eq. (9) when $\mathbf{R}_{T\mathbf{H}_i}^B \approx \mathbf{R}_{T\mathbf{H}_j}^B$. In this case, the effect of the estimate of $\mathbf{t}_{S\mathbf{H}}$ is greatly reduced.

It is helpful for solving the rotation matrix. Moreover, the estimate error of $\mathbf{t}_{S\mathbf{H}}$ can be isolated.

However, at the same time, when a pair of feature points is selected with similar positions in the sensor coordinate system (i.e., $\mathbf{p}_{Si} \approx \mathbf{p}_{Sj}$), the first part of the equation becomes meaningless ($\mathbf{R}_{T\mathbf{H}_i}^B \mathbf{R}_{S\mathbf{H}}^\dagger \mathbf{p}_{Si} \approx \mathbf{R}_{T\mathbf{H}_j}^B \mathbf{R}_{S\mathbf{H}}^\dagger \mathbf{p}_{Sj}$). This means that the equation is mainly affected by robot positioning errors and reconstruction errors at feature points rather than by the unknown variable $\mathbf{t}_{S\mathbf{H}}$, and the solution becomes invalid. To avoid this situation, a parallel judgement is presented to refine the calibration data. All involved pairs need to satisfy either of the following criteria:

$$\begin{cases} \|\mathbf{p}_{Si} - \mathbf{p}_{Sj}\| > d_{\parallel} & , \|\log(\mathbf{R}_{T\mathbf{H}_i}^{-1} \mathbf{R}_{T\mathbf{H}_j}^B)\| < \phi \\ \|\mathbf{p}_{Si} - \mathbf{p}_{Sj}\| > d_0 & , \|\log(\mathbf{R}_{T\mathbf{H}_i}^{-1} \mathbf{R}_{T\mathbf{H}_j}^B)\| \geq \phi \end{cases} \quad (17)$$

where $\log(\mathbf{R})$ represents the logarithm mapping from $SO(3)$ to $so(3)$.

The value of $\|\log(\mathbf{R}_{T\mathbf{H}_i}^{-1} \mathbf{R}_{T\mathbf{H}_j}^B)\|$ is used to determine whether the pair of poses can be approximately regarded as parallel according to a threshold ϕ . When $\|\log(\mathbf{R}_{T\mathbf{H}_i}^{-1} \mathbf{R}_{T\mathbf{H}_j}^B)\| < \phi$, the estimate error $\mathbf{t}_{S\mathbf{H}}$ is less involved in iterative operations, but a more stringent standard $d_{\parallel}$ is used to refine the involved pairs of calibration data. When $\|\log(\mathbf{R}_{T\mathbf{H}_i}^{-1} \mathbf{R}_{T\mathbf{H}_j}^B)\| \geq \phi$, the robot poses involved are likely to be a pair of random poses, and the involved pairs of calibration data need to meet only minimal conditions d_0 , which can be determined by the visual reconstruction error.

It is worth noting that the proposed solution is not limited by the robot measurement strategy because arbitrary poses are involved in the iterative process and only a small part of the measured data is eliminated by parallel judgement. Furthermore, when $\mathbf{R}_{T\mathbf{H}_i}^B = \mathbf{R}_{T\mathbf{H}_j}^B$ is guaranteed by a special movement strategy, the performance of the solution can be improved, as shown by the results of the previous analysis. The strategy can be implemented simply by translating the robot end when measuring the targeted plane, and its performance is assessed in the simulation section.

Finally, considering Eq. (9), the estimate of $\mathbf{n}_B$ has a large effect on the iteration, and thus, it needs to be updated as quickly as possible. In each iterative loop, once $\mathbf{R}_{S\mathbf{H}}^\dagger$ is obtained by Eq. (15), each point on the laser stripe is recalculated, and $\widehat{\mathbf{n}}_B$ is updated by Eqs. (3)–(7) before solving for $\mathbf{t}_{S\mathbf{H}}^*$. This additional updating works to accelerate convergence and increase the stability of the solution $\mathbf{t}_{S\mathbf{H}}^*$.

2.4. Initial value optimization

The proposed parallel judgement increases the stability in terms of data refinement, but this is not enough. First, the judgement is actually a fuzzy rule and is used only to remove data that are obviously not calculable. Moreover, other bad data from the measurement strategy and robot positioning errors are difficult to recognize by an artificially defined standard. Second, in the case of small amounts of data, rigorous data refining does more harm than good. Moreover, the unforeseen

Table 1

Parameters of simulated annealing.

Parameters	Nomenclature	Value
Initial temperature	$Temp_0$	50
Lowest temperature	$Temp_{lowest}$	0.02
Cooling ratio	k	0.5
Iterative times	$times$	1000

initial value greatly affects the iterative solution, and initial values with large deviations may even cause iterations to not converge.

It is necessary to apply an auxiliary algorithm in terms of initial value optimization. Thus, a fast simulated annealing algorithm is utilized to optimize the calibration process. The benefits of implementing simulated annealing to improve the proposed iterative method are obvious. On the one hand, simulated annealing is good at approximating the global optimization in a fixed amount of time, and it can be adopted to quickly optimize initial values when exceptions occur during the iteration due to local optimization and measuring errors. On the other hand, the difficulty of obtaining a precise optimum in simulated annealing can also be compensated for by the iterative method because simulated annealing is used only to roughly optimize the initial value in harsh situations (e.g., initial values with large deviations, large measurement errors and lack of data).

To adapt the proposed iterative method, a fast simulated annealing algorithm is designed in this paper. The main modifications are rapid cooling and global memory. Rapid cooling ensures that the simulated annealing process is roughly optimized as fast as possible because a more precise optimum from simulated annealing is unnecessary for the proposed iterative method and is time-consuming. Thus, a group of special parameters with very small cooling ratio (nearly 0.5) can be set for achieving rapid cooling simulated annealing, which is much faster than the original algorithm (normal cooling ratio is over 0.95), and other parameters can also be adjusted to accelerate the process. For example, the parameters in the table are used for subsequent simulations and experiments (see Table 1).

Global memory has been introduced as a solution to the above-mentioned problems caused by rapid cooling. At high temperatures, a better solution has a high possibility of being deprecated. Because of rapid cooling, it is likely that the algorithm will reach a low temperature before implementing effective optimization. To guarantee the effectiveness of the optimization, global memory is used to record the best optimum in each temperature equilibrium process, which is then inputted as the initial value for the next temperature.

For other details of the algorithm, a random technique based on the Euler angle is used to pick the neighbour of the given value.

$$\widehat{\mathbf{t}}_{S\mathbf{H}}^{new} = \begin{bmatrix} \mathbf{R}_{rand} \mathbf{R}_{S\mathbf{H}}^\dagger \mathbf{t}_{S\mathbf{H}}^{temp} + \mathbf{t}_{rand} \\ 0 & 1 \end{bmatrix} \quad (18)$$

where

$$\mathbf{R}_{rand} = Rot(\omega_r, \theta)$$

$$\mathbf{t}_{rand} = l \cdot \omega_t$$

$$\theta \sim U(-rlim, rlim)$$

$$l \sim U(0, tlim)$$

The form $Rot(\omega, \theta)$ represents the rotation matrix, which rotates the angle θ around the axis ω . ω_r and ω_t are random unit 3-D directions, and l is the length of $\mathbf{t}_{rand}$. The bounds of the uniform distributions for random rotation and translation ($rlim, tlim$) can be set to be proportional to temperature, which reduces redundant searches at low temperature.

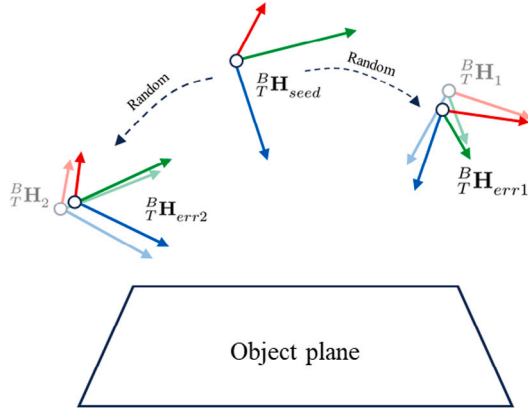


Fig. 4. Random technique for measuring poses generation.

In this paper, planar reconstruction accuracy (PRA) is introduced to evaluate $\hat{T}_S^B \mathbf{H}$ in simulated annealing. This indicator is strongly correlated with the performance of $\hat{T}_S^B \mathbf{H}$, which is verified in the simulation section.

$$f_{PRA} = \frac{\sum_{i=1}^n (\mathbf{p}_{Bi} - \frac{\sum_{i=1}^n \mathbf{p}_{Bi}}{n}) \cdot \mathbf{n}_B}{n} \quad (19)$$

The points $\mathbf{p}_{Bi}$ and $\mathbf{n}_B$ are obtained by Eqs. (3)–(7) by updating the estimate $\hat{T}_S^B \mathbf{H}$.

The modified simulated annealing algorithm is given below.

Algorithm 1 Fast simulated annealing

Input: initial value $\hat{T}_S^B \mathbf{H}_0$

- 1: Initialization: $Temp_0, Temp_{lowest}, f_{end}, k, times, \hat{T}_S^B \mathbf{H}_m \leftarrow \hat{T}_S^B \mathbf{H}_0$
- 2: calculate f_0 based on Equation (19)
- 3: $f_{global} \leftarrow f_0, f_{local} \leftarrow f_0, f_{new} \leftarrow f_0$
- 4: **while** $Temp > Temp_{lowest}$ **do**
- 5: $\hat{T}_S^B \mathbf{H}_{temp} \leftarrow \hat{T}_S^B \mathbf{H}_m$
- 6: **for** $i = 0$ **to** $times, f_{new} > f_{end}$ **do**
- 7: Obtain $\hat{T}_S^B \mathbf{H}_{new}$ from Equation (18)
- 8: Update f_{new} based on Equation (19)
- 9: **if** $f_{new} < f_{local}$ **then**
- 10: $f_{local} \leftarrow f_{new}$
- 11: $\hat{T}_S^B \mathbf{H}_{temp} \leftarrow \hat{T}_S^B \mathbf{H}_{new}$
- 12: **if** $f_{new} < f_{global}$ **then**
- 13: $f_{global} \leftarrow f_{new}$
- 14: $\hat{T}_S^B \mathbf{H}_m \leftarrow \hat{T}_S^B \mathbf{H}_{temp}$
- 15: **end if**
- 16: **else if** $e^{\frac{f_{local} - f_{new}}{Temp}} > p \in U(0, 1)$ **then**
- 17: $f_{local} \leftarrow f_{new}$
- 18: $\hat{T}_S^B \mathbf{H}_{temp} \leftarrow \hat{T}_S^B \mathbf{H}_{new}$
- 19: **end if**
- 20: **end for**
- 21: $Temp = k * Temp$
- 22: **end while**

Output: $\hat{T}_S^B \mathbf{H}_m$

3. Simulation

3.1. Initial value and injected error

In the simulation section, the proposed hand-eye calibration method is evaluated with different initial values and measurement errors. To generate the data inputted from the simulation measurements, the real

Table 2

Parameters of the noise distributions.

Noise levels	θ_{noise}	l_{noise}
Low	$U(-0.025^\circ, 0.025^\circ)$	$U(-0.1 \text{ mm}, 0.1 \text{ mm})$
Medium	$U(-0.1^\circ, 0.1^\circ)$	$U(-0.25 \text{ mm}, 0.25 \text{ mm})$
High	$U(-0.25^\circ, 0.25^\circ)$	$U(-1 \text{ mm}, 1 \text{ mm})$

values of the hand-eye matrix $T_S^B \mathbf{H}_0$ and the target plane are determined first. Then, according to an initial pose $B_T \mathbf{H}_{seed}$, a group of measuring poses are generated by a random technique, as in Fig. 4 and Eq. (20).

$$B_T \mathbf{H} = \begin{bmatrix} \mathbf{R}_{T_{seed}^B \mathbf{H}} \mathbf{R}_{adjust} & \mathbf{t}_{T_{seed}^B \mathbf{H}} + l_{rand} \cdot \omega_t \\ 0 & 1 \end{bmatrix} \quad (20)$$

$$l_{rand} \sim U(0, 150) \text{ (mm)}$$

In the equation, ω_t is a random unit 3-D direction. $l_{rand} \cdot \omega_t$ is used to randomly generate new measurement positions, while $\mathbf{R}_{adjust}$ is used to adjust the robot posture so that the stripes on object plane can be captured by the camera, and finally obtain new measuring poses ($B_T \mathbf{H}_1$ and $B_T \mathbf{H}_2$ in Fig. 4). The upper limit of the length l_{rand} is set to simulate rough mounting and measurement errors.

The points $\mathbf{p}_S$ that are measured by the structured-light vision sensor can be calculated as follows:

$$\begin{bmatrix} \mathbf{p}_S \\ 1 \end{bmatrix} = T_S^B \mathbf{H}_{real}^{-1} B_T \mathbf{H}^{-1} \begin{bmatrix} \mathbf{p}_B \\ 1 \end{bmatrix} \quad (21)$$

For the robot visual measurement system, the robot positioning error (nearly 1 mm) [24] is much more dominant than the feature point measuring error in the images (less than 0.1 mm). For simplification, only the robot positioning error is considered below, and the involved poses with errors ($B_T \mathbf{H}_{err1}$ and $B_T \mathbf{H}_{err2}$ in Fig. 4) are generated by the same technique as Eq. (18). The parameters of the random distributions in Eq. (22) are shown in Table 2. The high level of noise is set to simulate real-world scenarios.

$$B_T \mathbf{H}_{err} = \begin{bmatrix} \mathbf{R}_{noise} \mathbf{R}_{T^B \mathbf{H}} & \mathbf{t}_{T^B \mathbf{H}} + \mathbf{t}_{noise} \\ 0 & 1 \end{bmatrix} \quad (22)$$

where

$$\mathbf{R}_{noise} = Rot(\omega_r, \theta_{noise})$$

$$\mathbf{t}_{noise} = l_{noise} \cdot \omega_t$$

3.2. Parametric analysis

Before simulating the proposed calibration method and analysing its performance, it is necessary to discuss several parameters that have a significant impact on the calibration results. In the simulated annealing algorithm, the planar reconstruction accuracy is used to evaluate the initial value optimization. Fig. 5 shows the iteration trends of 100 simulations with high noise level in Table 2. As shown, the rotation errors and the translation error have similar trends, as does the indicator PRA. It is verified that the planar reconstruction accuracy is sufficient for evaluating the accuracy of the results and predicting iterative anomalies. Fig. 6 shows how the fast simulated annealing based on f reverses an unsuccessful simulation calibration. The performance of the initial value optimization by simulated annealing is significant. The parameters for rapid cooling achieve rapid optimization in large deviation and global memory, ensuring effectiveness in a finite number of iterations.

The number of measurements is a main factor that affects the calibration results and efficiency of the real-world calibration process. Fig. 7 shows the hand-eye calibration results for different numbers of scanned lines with high noise level. As shown, calibrations with fewer than 90 measurements are likely to be unstable because of poor

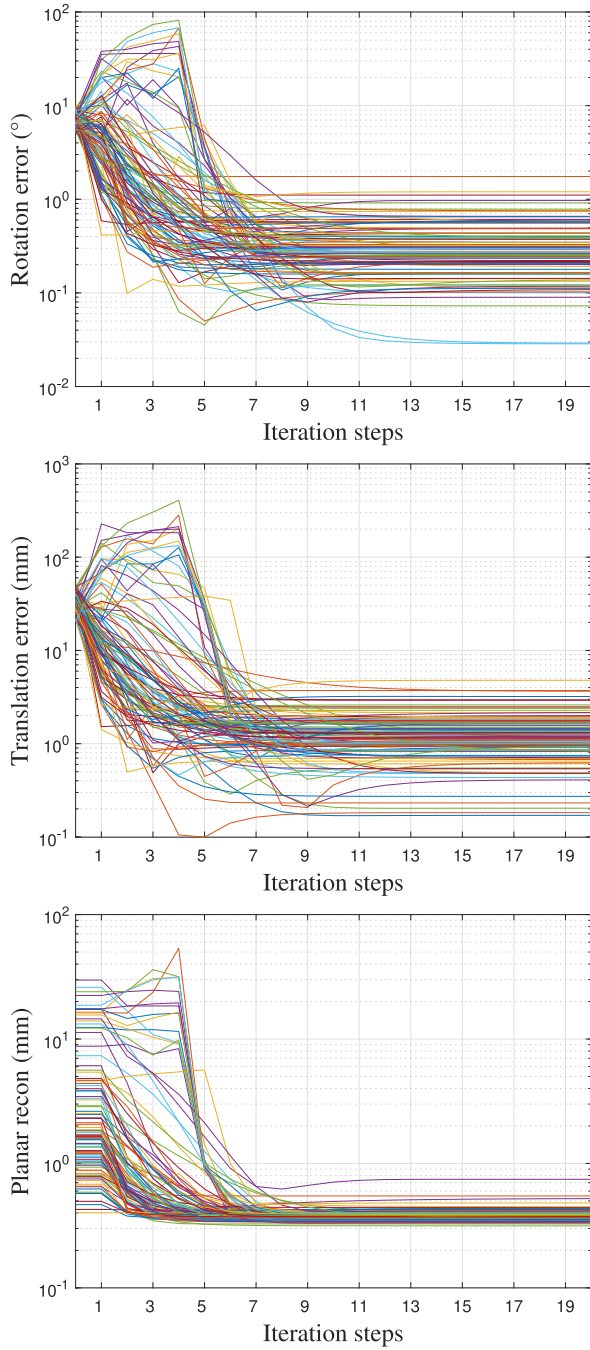


Fig. 5. The trends of the rotation error, translation error and planar reconstruction accuracy with 100 simulations within 20 iterations.

calibrated data. Calibration with 120 measurements is significantly better than calibration with fewer measurements. At more than 120 measurements, the error distributions did not change significantly for either the rotation or translation part, but the number of measurements greatly increased. To balance the calibration accuracy and efficiency, calibrated data from 120 measurements were adopted in subsequent simulations and experiments.

3.3. Results

The convergence process of the calibration results is illustrated in Fig. 8.

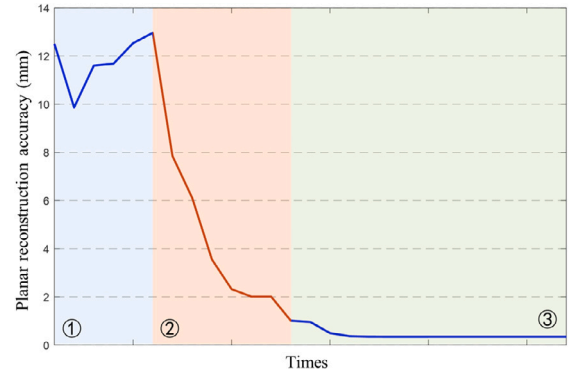


Fig. 6. The fast simulated annealing based on f prevents unsuccessful simulation calibration. 1. Iterative anomaly recognition. 2. Simulated annealing process. 3. Two-step iteration process.

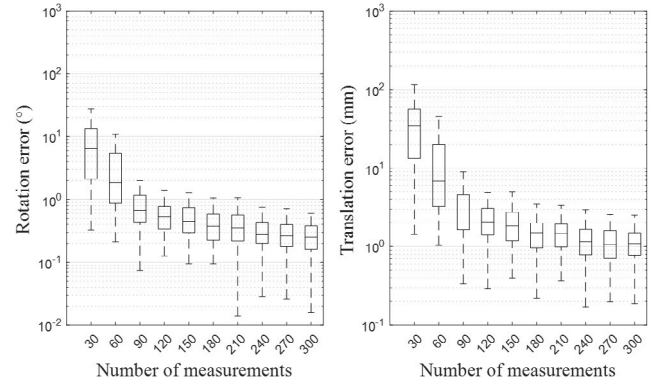


Fig. 7. Simulated results with different quantities of scanned lines.

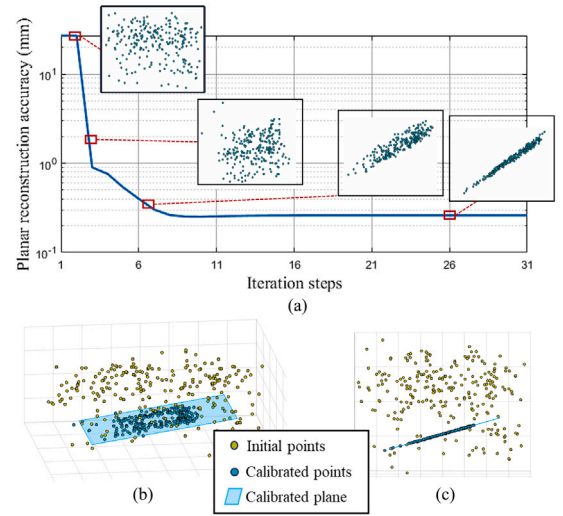


Fig. 8. The convergence process of the calibration results. (a) As the calibration error decreases, the distribution of the reconstructed points on the laser stripes converges to a plane. (b) and (c) show the reconstruction results based on the initial values and the final results, respectively.

Figs. 9–11 illustrate the results of 100 trials with different initial values and noise levels, and the abscissa of 0 represents the initial value. Table 3 shows the difference level distributions of the initial values, which are generated similarly to those of the noise injection.

Table 3
Parameters of the initial value distributions.

Deviation levels	θ_d	l_d
Low	$N(1^\circ, 5^\circ)$	$N(10 \text{ mm}, 20 \text{ mm})$
Medium	$N(5^\circ, 10^\circ)$	$N(20 \text{ mm}, 50 \text{ mm})$
High	$N(10^\circ, 20^\circ)$	$N(50 \text{ mm}, 100 \text{ mm})$

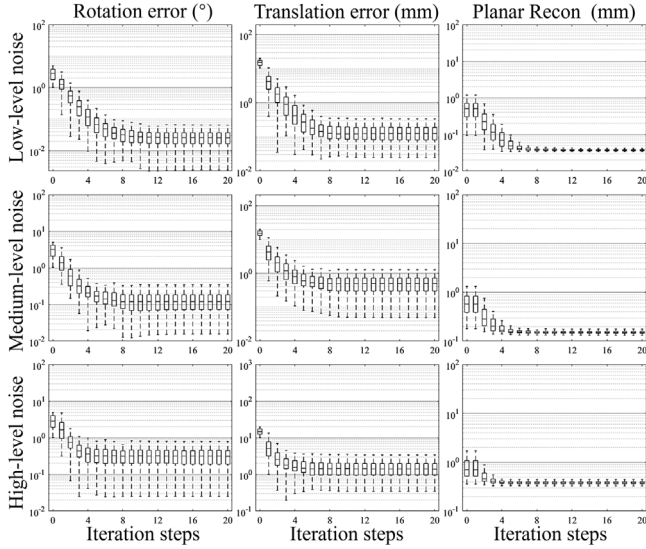


Fig. 9. Simulation results under low-level initial values obtained by injecting different noise levels.

Normal distributions are used to generate initial values from the real values. Box plots are used to describe the statistical characteristics of the results, which can reflect the stability of the proposed method. There is a clear correlation between the decrease in the rotation error and translation error, which is due to the update $\hat{n}_B$ before evaluating $t_{S,H}^*$. All the results demonstrate fast convergence and stabilization within 10 iterations.

In each figure that has the same initial value input, the iterative results have the same trend of decreasing evaluation error with different noise levels. As shown, the convergence efficiency is likely unaffected by the difference in noise levels. The noise levels affect only the final stable results, which are roughly distributed around the injected noise.

With increasing initial deviations, the number of iterations needed to achieve convergence increases. With a large initial deviation, both the rotation and translation errors fluctuate greatly at the start of iteration. However, the errors quickly decrease after the fifth iteration because the simulated annealing takes effect to avoid unsuccessful iteration. The figure shows that simulated annealing significantly optimizes large initial deviations, which improves the stability of the method.

4. Experiment

4.1. System setup

Compared to simulations, unpredictable noise in the real world is likely to affect the performance of the proposed method. To verify the robustness and stability of the proposed method, experiments are carried out on a robot vision system with a structured light sensor. As shown in Fig. 12, the calibration involved a 6-axis Yaskawa Motoman MH12 robot (repeatability $\pm 0.08 \text{ mm}$), a StingRay 660 nm line structured-light emitter and an AVT Mako G-125 CCD camera (1292×964) with a Kowa LM2JC 12 mm lens. The emitter and camera are integrated and fixed on the robot end effector.

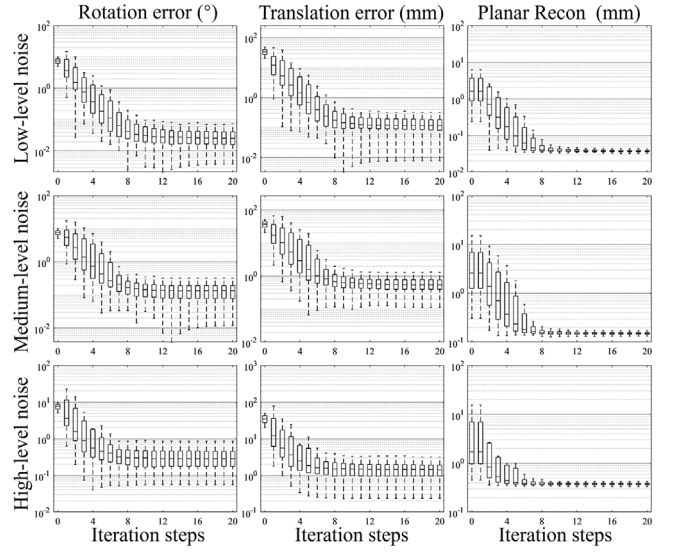


Fig. 10. Simulation results under medium-level initial values obtained by injecting different noise levels.

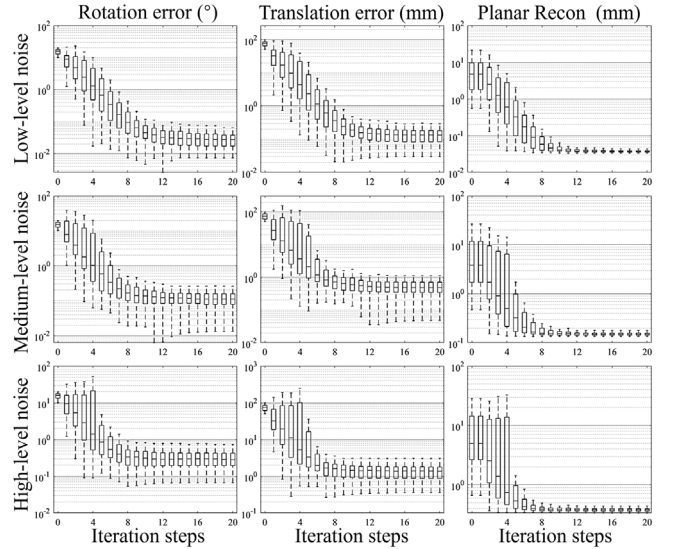


Fig. 11. Simulation results under high-level initial values obtained by injecting different noise levels.

To evaluate the stability of the proposed method, a measurement strategy with random poses is implemented for the experiment. The values of the robot poses are directly read from the robot controller DX200, and the corresponding laser stripe images are processed via centreline extraction.

4.2. Calibration procedure

Experiments implement a complete calibration process to verify the performance of the proposed method. The procedure is divided into four steps: measurement, detection, initial optimization, and iterative solution. During the measurements, the laser line stripes on the object plate are manually captured with random poses. If the plane can be positioned easily, the measurement process can be automatically performed with a rough initial hand-eye relationship. An adaptive Laplacian Gaussian filter is used to decrease noise and recognize the points on the laser stripes. Due to the uncertainty of the sensor noise, the points on the laser stripe are not used directly in the subsequent

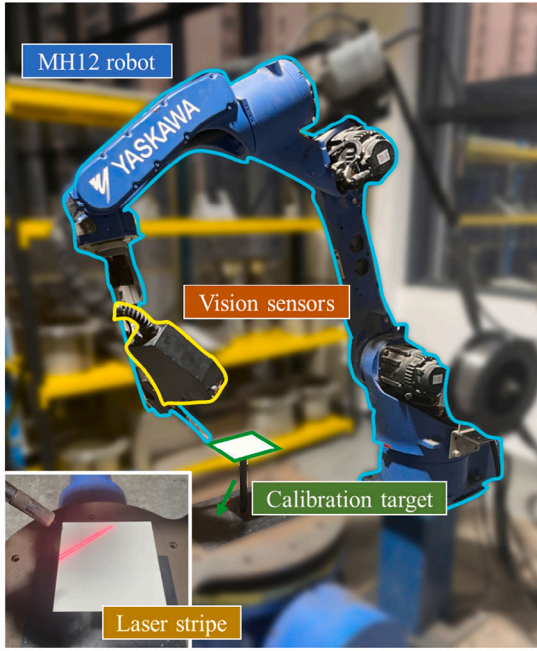


Fig. 12. The experimental scene.

Table 4
Experiments result.

	Planar reconstruction accuracy	
	Initial (mm)	Final (mm)
Exp. A	26.5829	0.3496
Exp. B	27.0093	0.2606
Exp. C	19.3171	0.2687
Exp. D	29.4238	0.2764
Exp. E	22.9506	0.3263

calculation. All the points extracted on the laser stripe are first fitted to a straight line, and the points on the fitted line are involved in the iterations.

Since there is no ground truth, the planar reconstruction accuracy is used to evaluate the accuracy of the results after calibration. The planar reconstruction accuracy reflects the average distance from points to planes, and its evaluation performance has been proven in the simulation section.

4.3. Results

4.3.1. Repeated experiments

The method performance is verified under different environments through experiments conducted in five cases. In the measuring process, a ceramic plate with a machining accuracy of 0.05 mm is used as the calibration target (Fig. 12). In those experiments, the ceramic plate were fixed at different positions, and different measuring poses were obtained from the random pose generator. Each experiment involve 120 measuring poses, and two points on each stripe are used for iteration. In order to assess the deviation of the initial error, a relatively precise initial value is acquired initially by the hand-eye calibration utilizing a dedicated calibration plate. Then, similar to the simulation, high-level deviation are injected into the initial value before the iteration.

Fig. 13 and Table 4 show the results and iterative process of repeated experiments. As the results shown, the convergence behaviour is similar, and the final errors are small enough. In addition, the results of Exp. A and E are slightly unsatisfactory compared to other experiments due to the generation of inadequate measuring poses.

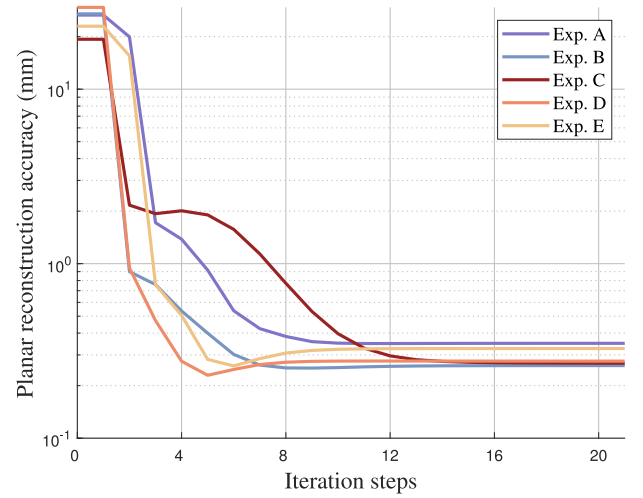


Fig. 13. The repeated experiments result.

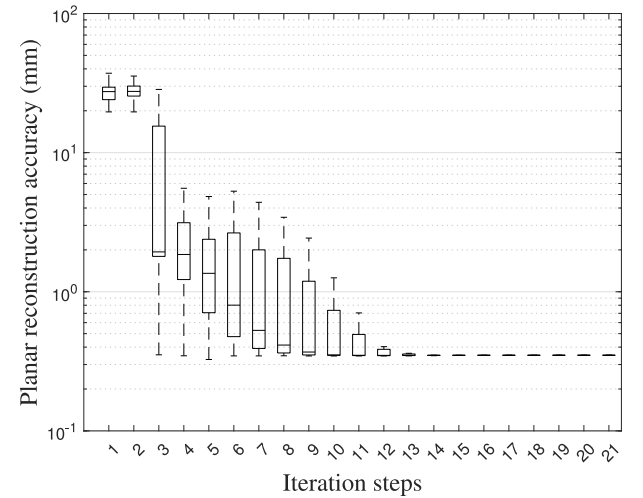


Fig. 14. The convergence stability of the method with different initial values.

4.3.2. Stability experiments

Due to the uncertainty in real calibration scenes, it is necessary to evaluate the method stability for different initial values in a same case. Two hundred initial values with high-level injected error distribution are inputted to the 120-pose experimental data to solve the calibration problem. The convergence results are illustrated in Fig. 14. As the results shown, different initial values converge to a unified value within 20 iterations.

4.3.3. Comparative experiment on workpieces

To analyse the method accuracy, traditional hand-eye calibrations with dedicated calibration plates (Fig. 15(a)) are also implemented for the comparisons with several indicators. Additionally, the proposed method is not limited to special calibration objects, and any machined plane can be used for calibration. This experiment implements line-structured light eye-in-hand calibration on a common workpiece that has a flat surface (shown in Fig. 15).

Reprojected errors: Commonly, reprojected rotation errors $\mathbf{R}_{err}$ and translation errors $\mathbf{t}_{err}$ are utilized to estimate the accuracy of hand-eye calibration [25]. However, neither of these indicators applies to structured light hand-eye calibration based on planar constraints due to the lack of image features. Therefore, an additional set of calibration data with dedicated calibration plates (A_{test} and B_{test}), the same as

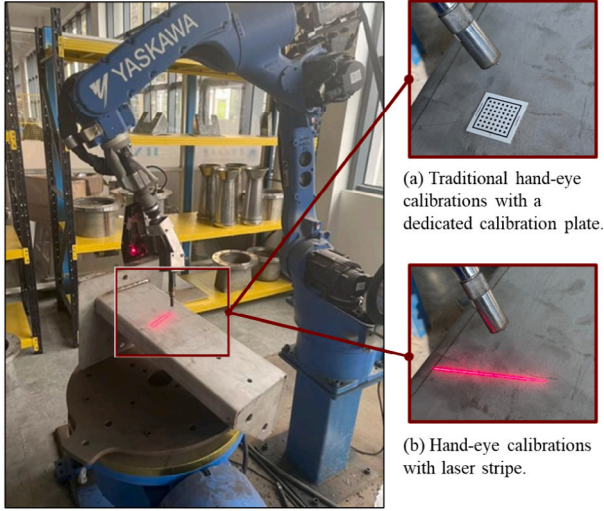


Fig. 15. Two kinds of hand-eye calibration are implemented on a common workpiece, both using the same calibrated plane.

traditional methods, is collected to calculate the reprojected error with structured light hand-eye calibration results $\hat{\mathbf{X}}$ based on the equations:

$$\mathbf{Y} = \mathbf{A}_{\text{test}} \hat{\mathbf{X}} \mathbf{B}_{\text{test}} \quad (23)$$

$$\mathbf{R}_{\text{err}} = \sqrt{\frac{1}{n} \sum_{i,j,i \neq j} \|\log(\mathbf{R}_{Y_i}^T \mathbf{R}_{Y_j})\|^2} \quad (24)$$

$$\mathbf{t}_{\text{err}} = \sqrt{\frac{1}{n} \sum_{i,j,i \neq j} \|\mathbf{t}_{Y_i} - \mathbf{t}_{Y_j}\|^2} \quad (25)$$

where $\log(\mathbf{R})$ represents the logarithm mapping from $SO(3)$ to $so(3)$.

Sphere fitting: To compare the calibration results under the same criteria, a sphere fitting experiment is also used to evaluate the results from traditional methods. After implementing traditional calibrations with a dedicated calibration plate, the results are used to reconstruct a precision sphere with a 12.5 ± 0.005 mm radius. The process of the reconstruction experiment is illustrated in Fig. 16. The laser emitter projects stripes on a ceramic ball with 6 robot rotation groups and 10 translations in each group, which yields 60 stripe images for sphere fitting. After stripe detection via image processing and point cloud reconstruction via hand-eye calibration, the sphere can be fit, and its radius reflects the accuracy of the result.

Planar reconstruction accuracy: This indicator is also used to evaluate the performance of traditional methods for comparison. Illustrated in Fig. 15, all calibrations are implemented on the same plane, which will be reconstructed by different hand-eye results.

Traditional methods [2–7] are implemented for comparative experiments, and the results are evaluated by reprojected errors, sphere fitting results and planar reconstruction accuracy (Table 5). As the results shown, the proposed method based on planar constraints is capable of achieving similar rotation accuracy and slightly poor translation accuracy compared with other traditional calibration methods. Despite using only a planar reference rather than dedicated calibration plates, the sphere fitting performance of the proposed method is little worse ($\approx 1.5\%$) than that of conventional methods. Driven by the optimization objective, the proposed method performs significantly better with regard to planar reconstruction accuracy, which facilitates follow-up robot machining on the calibrated surface (i.e., grinding, spraying, welding).

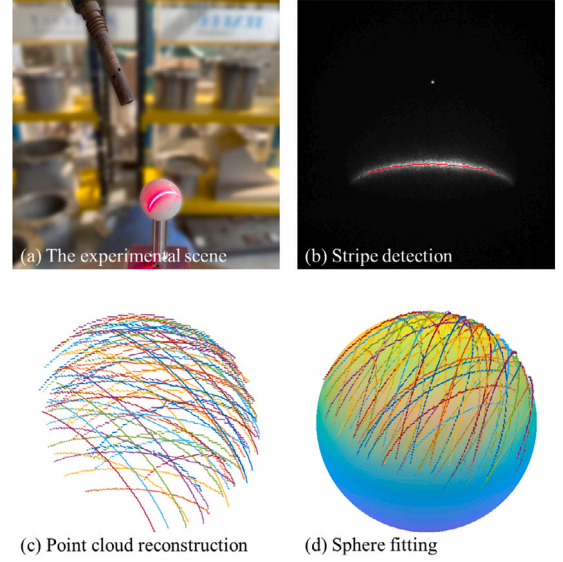


Fig. 16. The process of the reconstruction experiment.

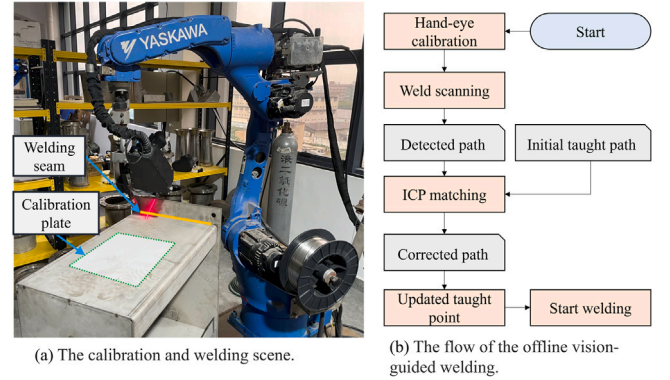


Fig. 17. The offline vision-guided welding experiment scene and process flow.

4.3.4. Application experiment of vision-guided welding

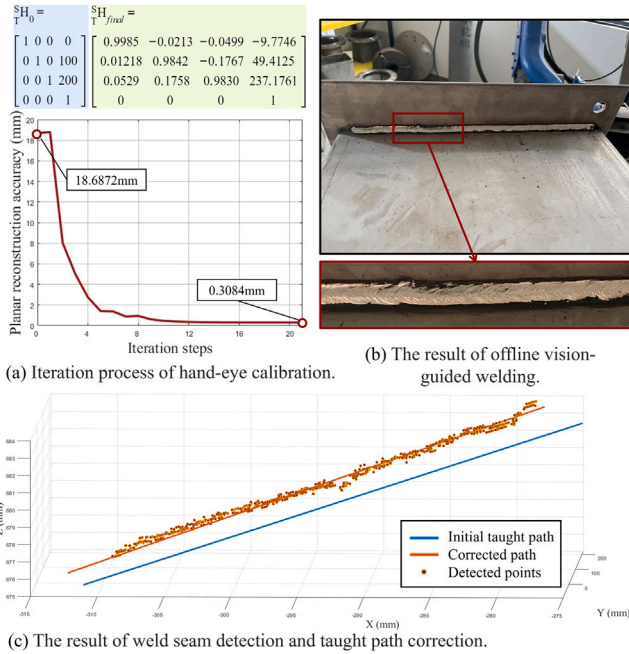
An offline vision-guided welding experiment is implemented as the application case verification of the proposed method. The welding machine MOTOWELD-RD350 is used in this experiment to perform MIG welding. As shown in Fig. 17(a), the robot firstly performs the proposed hand-eye calibration method on a workpiece surface near the weld seam. In the experiment, measuring motion have been simply programmed and run automatically, and the measuring take about 5–10 s for each pose. And then the calibration result is used in the offline welding process. The vision-guided welding process is shown in Fig. 17(b). The robot scans the weld seam with the structured light vision sensor firstly and obtains the detected path. The 3D detected points on the path is acquired by the conversion from the sensor frame to the base frame (Eq. (2)). Then the detected path is matched with the initial taught path which has large positioning deviation, and corrected taught path can be obtained based on the matching result. Finally, the robotic welding can be performed after updating the corrected taught points to the robot controller. More details about the offline welding are introduced in paper [26].

The results of calibration and the offline welding are shown in Fig. 18(a). The synthetic data ${}^S_T\mathbf{H}_0$ is set as initial hand-eye transformation matrix, which has large planar reconstruction error (18.6872 mm). After the iteration process, reconstruction error decreases greatly and the final calibration result ${}^S_T\mathbf{H}_{\text{final}}$ has only 0.3084 mm reconstruction error. The final calibration result is used to convert the detected points

Table 5

Comparison of the reprojected errors, sphere fitting results and planar reconstruction accuracy.

Methods	Rotation errors	Translation errors	Fitting radius (Relative errors)	Planar recon
Park [3]	0.7301°	1.0073 mm	12.7231 (+1.7848%) mm	0.9352 mm
Tsai [2]	0.7301°	1.0077 mm	12.7241 (+1.7928%) mm	0.9342 mm
Horaud [4]	0.7301°	1.0074 mm	12.7226 (+1.7808%) mm	0.9354 mm
Daniilidis [5]	0.7203°	0.7396 mm	12.6296 (+1.0371%) mm	1.0842 mm
Shah [6]	0.6435°	0.8814 mm	12.8786 (+3.0285%) mm	0.8368 mm
Ha [7]	0.6606°	0.8723 mm	12.5822 (+0.6575%) mm	1.0289 mm
Paper	0.7385°	2.0013 mm	12.9201 (+3.3608%) mm	0.3262 mm

**Fig. 18.** The hand-eye calibration result and the offline robotic welding results.

to the base frame (18(c)), and the straight-line reconstruction root mean square error is 0.3027 mm. The weld shape of the workpiece after welding is shown in 18(b). It demonstrated that the calibration result from the proposed method has enough precision for vision-guided robotic welding applications.

4.4. Discussion

The proposed calibration procedure is implemented in real-world scenarios to validate its performance in this section.

Convergence behaviour: Repeated experiments results show stability convergence behaviour of the 5 iterations, and all the cases can achieve a small planar reconstruction error. The result also indicates that the quality of the pose distribution can affect the results of the solution, suggesting that customized distributions may further optimize the calibration results.

Stability: The result of stability experiments also demonstrate the stable and rapid convergence of the proposed method. Stability experiments confirm that the solution adapts well to different initial values, even with high-level errors, thereby validating the reliability of the method.

Accuracy: Accuracy experiments are conducted by comparing it with traditional solutions using three different criteria. As the results shown, the proposed method can achieve similar evaluation with reprojected errors and fitting radius, and its planar reconstruction error is lower than others. The calibration result is also used in application experiment, and vision-guided welding results verify the accuracy of the proposed method.

Limitations: While the calibration results meet application requirements and achieve high planar reconstruction accuracy, they exhibit poorer performance in reprojected errors and sphere fitting compared to conventional calibrations. This can be attributed to two main reasons: planar reconstruction index is used as the optimization goals of our iteration instead of traditional error index, and commonly machined surfaces is used for calibration rather than specialized calibration targets with high precision.

To sum up, these experimental results substantiate that the proposed method can effectively facilitate hand-eye calibration implementation without special calibration objects, simplifying automation processes and enabling precision maintenance for robotic manufacturing.

5. Conclusion

In this paper, a hand-eye calibration method for a line-structured light eye-in-hand robot vision system based on planar constraints is proposed. The aim of this method is to improve the applicability and stability of the calibration procedure. Moreover, this approach can be applied to arbitrary measuring strategies with a single reference plane. Line-structured light vision with an eye-in-hand configuration is commonly used in robotic manufacturing. Thus, it is feasible to implement hand-eye calibration directly on workpieces with flat surfaces.

A two-step iterative method is implemented, and the rotation and translation steps are sequentially solved based on the initial values and planar constraints. The data quality is considered after the measurements are taken, and a parallel judgement is proposed based on the planar constraint equation to refine the calibration data. Because the initial value has a significant effect on the iteration, an initial value optimization based on simulated annealing is utilized to improve the method stability. Simulated calibrations can be implemented to evaluate the approach with different numbers of measurements, noise levels and initial deviations, and the results prove the stability of the approach. Real-world calibrations are also designed to assess the method performance, which shows that the proposed calibration procedure has good robustness and accuracy.

In future studies, special measuring strategies could be researched to reduce measurement costs and improve calibration accuracy. In addition, the combination of planar constraints with other geometric constraints can be further explored to improve the flexibility and stability of the method.

CRedit authorship contribution statement

Kaifan Zhong: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Jingxin Lin:** Writing – review & editing, Software, Investigation. **Tao Gong:** Writing – review & editing, Methodology, Investigation. **Xianmin Zhang:** Validation, Project administration. **Nianfeng Wang:** Writing – review & editing, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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